

Exercise 2: Neural Networks, Double Descent and Linear Regression

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Problem 1 (Bias–Variance via SVD):

Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ with $n < p$ and $\text{rank}(\mathbf{X}) = n$. Write the compact SVD $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ with singular values $\sigma_1, \dots, \sigma_n > 0$, and let $\eta \in \mathbb{R}^n$. Define the variance term

$$\mathcal{V}(\mathbf{X}, \eta) := \|\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}\eta\|_2^2.$$

1. Show that $\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1} = \mathbf{V}\mathbf{\Sigma}^+\mathbf{U}^\top$, where $\mathbf{\Sigma}^+$ is the Moore–Penrose pseudoinverse of $\mathbf{\Sigma}$.
2. Express $\mathcal{V}(\mathbf{X}, \eta)$ as a sum over the singular values of $\mathbf{X}$. Your final answer should have the form $\sum_{i=1}^n \frac{(u_i^\top \eta)^2}{\sigma_i^2}$, where u_i are the columns of $\mathbf{U}$.
3. If $\eta \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_n)$, compute $\mathbb{E}[\mathcal{V}(\mathbf{X}, \eta)]$ and relate it to $\text{tr}((\mathbf{X}\mathbf{X}^\top)^{-1})$.

Problem 2 (Leave-one-out via Sherman–Morrison):

Setup. Let $\mathbf{X} \in \mathbb{R}^{n \times p}$ be a design matrix with rows $\mathbf{x}_i^\top \in \mathbb{R}^{1 \times p}$ (data points) and columns (features), and assume $\text{rank}(\mathbf{X}) = p \leq n$ so that $\mathbf{X}^\top \mathbf{X}$ is invertible. Let $\mathbf{y} \in \mathbb{R}^n$ be the response vector. Define

$$\mathbf{A} := \mathbf{X}^\top \mathbf{X} \in \mathbb{R}^{p \times p}, \quad \hat{\mathbf{w}} := \mathbf{A}^{-1} \mathbf{X}^\top \mathbf{y} \in \mathbb{R}^p, \quad \mathbf{H} := \mathbf{X} \mathbf{A}^{-1} \mathbf{X}^\top \in \mathbb{R}^{n \times n}, \quad \hat{\mathbf{y}} := \mathbf{H} \mathbf{y} \in \mathbb{R}^n.$$

For $i \in \{1, \dots, n\}$, denote the i th leverage score by

$$\mathbf{h}_i := H_{ii} = \mathbf{x}_i^\top \mathbf{A}^{-1} \mathbf{x}_i \in [0, 1],$$

and let $\mathbf{X}_{-i} \in \mathbb{R}^{(n-1) \times p}$ and $\mathbf{y}_{-i} \in \mathbb{R}^{n-1}$ be $\mathbf{X}$ and $\mathbf{y}$ with row/entry i removed. The leave-one-out (LOO) estimator and its prediction at the removed point are

$$\hat{\mathbf{w}}^{(-i)} := (\mathbf{X}_{-i}^\top \mathbf{X}_{-i})^{-1} \mathbf{X}_{-i}^\top \mathbf{y}_{-i} \in \mathbb{R}^p, \quad \hat{\mathbf{y}}_i^{(-i)} := \mathbf{x}_i^\top \hat{\mathbf{w}}^{(-i)} \in \mathbb{R}.$$

Task. Note that $\mathbf{X}_{-i}^\top \mathbf{X}_{-i} = \mathbf{A} - \mathbf{x}_i \mathbf{x}_i^\top$. Apply the Sherman–Morrison rank-one inverse identity

$$(\mathbf{A} - \mathbf{u}\mathbf{v}^\top)^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1} \mathbf{u} \mathbf{v}^\top \mathbf{A}^{-1}}{1 - \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u}},$$

with $\mathbf{u} = \mathbf{v} = \mathbf{x}_i$, to $(\mathbf{X}_{-i}^\top \mathbf{X}_{-i})^{-1} = (\mathbf{A} - \mathbf{x}_i \mathbf{x}_i^\top)^{-1}$, and derive the leave-one-out residual formula

$$y_i - \hat{y}_i^{(-i)} = \frac{y_i - \hat{y}_i}{1 - h_i}.$$

Hint. First express $\hat{\mathbf{w}}^{(-i)}$ in terms of $\mathbf{A}^{-1}$, $\mathbf{x}_i$, $\mathbf{X}^\top \mathbf{y}$, and $\mathbf{y}_i$, then left-multiply by $\mathbf{x}_i^\top$ to obtain $\hat{\mathbf{y}}_i^{(-i)}$.

Problem 3 (Extreme singular values of Gaussian random matrices):

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be a random matrix with i.i.d. entries distributed as $\mathcal{N}(0, 1)$. Denote by $\sigma_{\min}(\mathbf{X})$ and $\sigma_{\max}(\mathbf{X})$ the smallest and largest singular values of $\mathbf{X}$, respectively, and define the *condition number*

$$\kappa(\mathbf{X}) = \frac{\sigma_{\max}(\mathbf{X})}{\sigma_{\min}(\mathbf{X})}.$$

1. For fixed d and varying n , generate n_{samples} independent Gaussian matrices $\mathbf{X}$ and compute the empirical averages of $\sigma_{\min}(\mathbf{X})$, $\sigma_{\max}(\mathbf{X})$, and $\kappa(\mathbf{X})$.
2. Compare your numerical results with the theoretical asymptotic predictions from random matrix theory:

$$\sigma_{\min}(\mathbf{X}) \approx \sqrt{n} - \sqrt{d}, \quad \sigma_{\max}(\mathbf{X}) \approx \sqrt{n} + \sqrt{d}, \quad \kappa(\mathbf{X}) \approx \frac{\sqrt{n} + \sqrt{d}}{\sqrt{n} - \sqrt{d}}.$$

3. Plot $\sigma_{\min}$, $\sigma_{\max}$, and κ as functions of n , and superimpose the theoretical curves.
4. Discuss how the empirical curves approach the theoretical predictions as n grows.

Hint: Use singular value decomposition (`np.linalg.svd`) to obtain $\sigma_{\min}$ and $\sigma_{\max}$.